

Sonny Alberto Medina

Data Scientist - PhD in Probability and Statistics

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PROFESSIONAL EXPERIENCE

Ford Motor Company, CDMX — *Data Scientist*

OCTOBER 2024 - NOW

Developing a multi-signal forecasting model to predict warranty claims.

Processed unstructured data using advanced techniques, including speech-to-text conversion, large language models (LLMs), and topic classification. Applied time series cross-correlation and feature engineering to enhance forecasting accuracy with exogenous data integration.

University of Bath, UK — *PhD Candidate /Teaching Assistant*

SEPTEMBER 2021 - OCTOBER 2024

PhD Candidate of the Statistical Applied Mathematics program.

2 papers accepted in top probability journals.

3 years as graduate teaching assistant in Probability & Statistics units.

4 think-tank consultancies: Roche, NU Mongolia, Schlumberger and Wessex Water.

Nielsen, Data Science - GDDC— *Data Scientist*

NOVEMBER 2018 - AUGUST 2020

Led the redesign of the Retail Establishments Survey for Mexico and Brazil.

Reducing, in the MX case a \$400K USD 10+ y.o. legacy fieldwork, operational costs by 75% and 3% of full cycle time, whilst meeting the sampling precision targets.

Human Rights Commission, CDMX — *Analyst*

AUGUST 2015 - AUGUST 2016

Automation of reports in R Markdown, creation of dashboards for describing the human rights observers' performance, and projects to reduce the delay on the resolution of the complaint records. Co-organisation of the first international congress of indicators in human rights.

SKILLS

Predictive modelling

Risk quantification

Time series forecasting

Bespoke Machine Learning

Modern off-the-shelf ML

Monte Carlo simulation

Survey sampling

Data/GIS visualisations

Natural language processing

GUI development

COMPUTER SKILLS

Python: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, PySpark, SciPy

R: Tidyverse, Plotly, Leaflet, Shiny, Survey

Excel: Pivot tables

SQL, SAS, Git, Google Cloud Platform.

LANGUAGES

Spanish: Native

English: Fluent (CEFR level C2)

EDUCATION

University of Bath, Bath, UK — *PhD Statistics*

SEPTEMBER 2020 - FEBRUARY 2025

Passed PhD viva and obtained 1st class honors MRes degree

CIMAT, Guanajuato — *MSc Probability & Statistics*

AUGUST 2016 - JUNE 2018

Graduated. GPA: 9.50/10.00

UNAM, Faculty of Sciences — *BSc Actuarial Science*

JULY 2011 - AUGUST 2015

Graduated with honours. GPA: 9.0/10.0

Theses

- PhD. [Excursions from hyperplanes for \$\alpha\$ -stable processes. Supervised by Prof. Andreas Kyprianou \(U. of Warwick\)](#)
- MSc. [Gerber-Shiu function for a renewal risk model with positive jumps and brownian perturbation.](#)
- BSc. [INEGI 2010 Census sample: variance estimation.](#)

RESEARCH PAPERS

[Time series models applied to the records of the Human Rights Commission of the Federal District](#)

[\$\alpha\$ -stable Lévy processes entering the half space or a slab](#)

[The Gerber-Shiu Penalty Function for a Two-sided Renewal Risk Process Perturbed by Diffusion](#)

ADDITIONAL INFORMATION

Talks

Excursions from hyperplanes for Lévy Processes, Turin-Bath PhD Workshop, University of Turin, Italy, 2023.

SARIMAX vs Bayesian Structural models, Bottles and Models Meetings, DSci Nielsen, 2021.

Estimation techniques in survey sampling with official data, Jr Seminar of Math, Univ. of Guanajuato, 2018.

Awards

Nielsen Simply Excellence Award. For the Retail Survey-design Review project, that implied a saving of more than \$30K US. and 60 work days for Data Acquisition through implementation of sample selection algorithms and KML map generation with R.

TEACHING

UNAM, FES Acatlan

AUGUST 2019 - MAY 2025

Lecturer:

Stochastic Processes I

Probability I

Risk Theory

CIMAT

JANUARY 2017 - DECEMBER 2017

Teaching assistant:

Data Analysis for Computer Sciences

Foundations of probability and Statistics

UNAM, Fac. of Sciences

AUGUST 2014 - JULY 2016

Teaching assistant:

Probability

Survey Sampling

Probability and Statistics for Computer Sciences